sas innovate 2026

Subflows in SAS Viya

Learning Lab

Grace Barnhill
Technical Training Consultant
Grace.Barnhill@SAS.com

Copyright © SAS Institute Inc. All rights reserved.

Copyright © SAS Institute Inc. All rights reserved.

Meet the Speaker

- Data Management Content Development
  SAS Education
- Located in Provincetown, MA
  (formerly at SAS HQ in Cary, NC)
- Basketball fan & crafter
  Let's chat about the Dallas Wings or cross stitching ☺

Copyright © SAS Institute Inc. All rights reserved.

sas innovate 2026

Copyright © SAS Institute Inc. All rights reserved.

sas innovate 2026

Agenda

SAS Studio Flows and Subflows Overview

Guided Hands-On (Building a Flow to use as a Subflow)

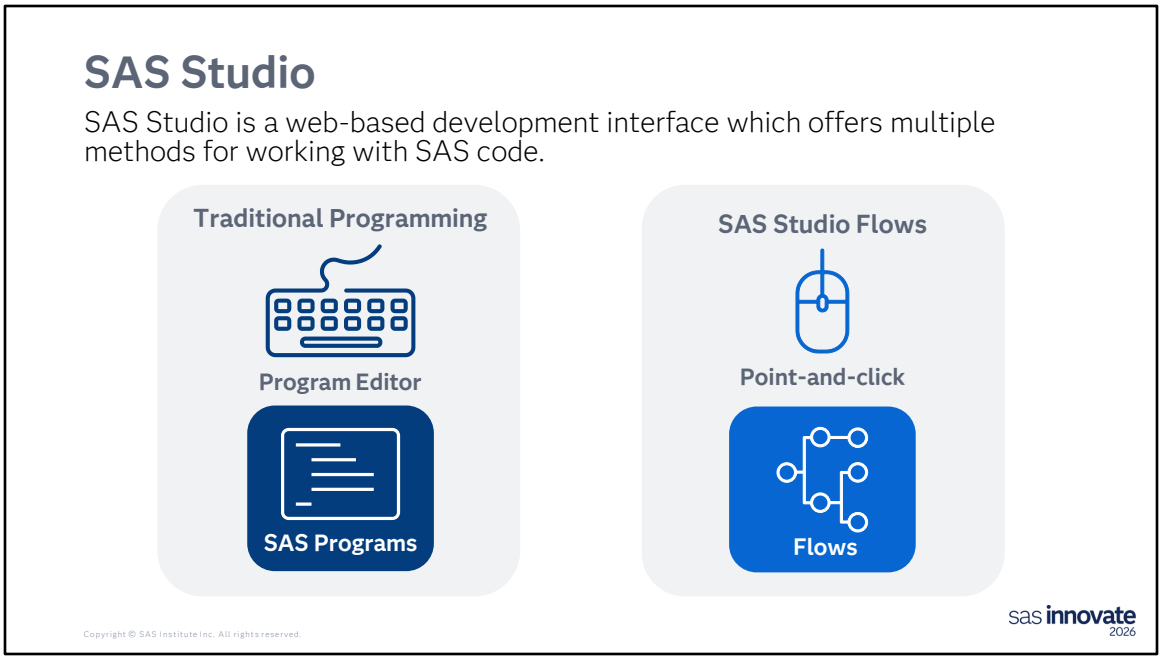
Questions & Closing

Copyright © SAS Institute Inc. All rights reserved.

sas innovate 2026

sas innovate 2026

Copyright © SAS Institute Inc. All rights reserved.



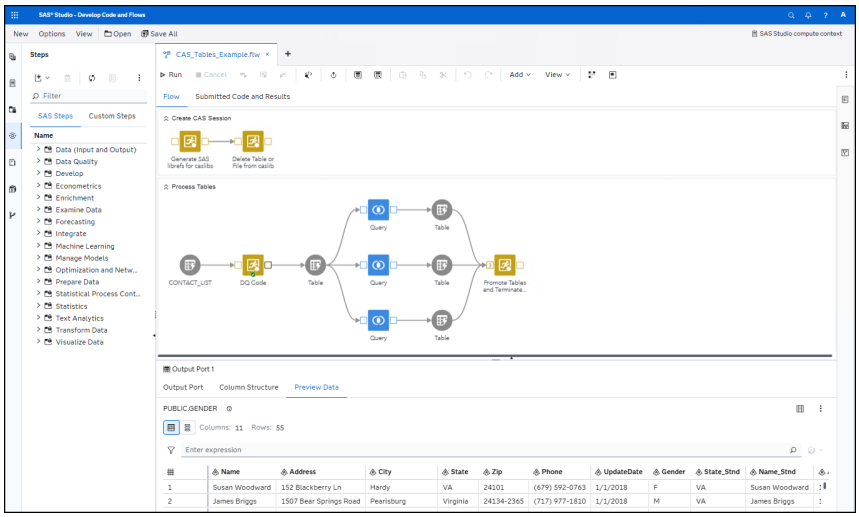
SAS Studio is a web-based development interface which offers multiple methods for working with SAS code.

The traditional method is to use the program editor to write SAS code and develop SAS programs.

Another method is to use pre-built point-and-click steps which generate SAS code for tasks such as accessing, analyzing, transforming, or visualizing data. These steps are connected in a dynamic flow chart, allowing you to visualize your data transformation process.

Copyright © SAS Institute Inc. All rights reserved.

SAS Studio – Flows



Copyright © SAS Institute Inc. All rights reserved.

sas innovate 2026

A *SAS Studio* **flow** is a sequence of operations on data. Data and operations are represented by nodes (steps). A flow orchestrates nodes in a series of steps in which the output of one node is the input to another node.

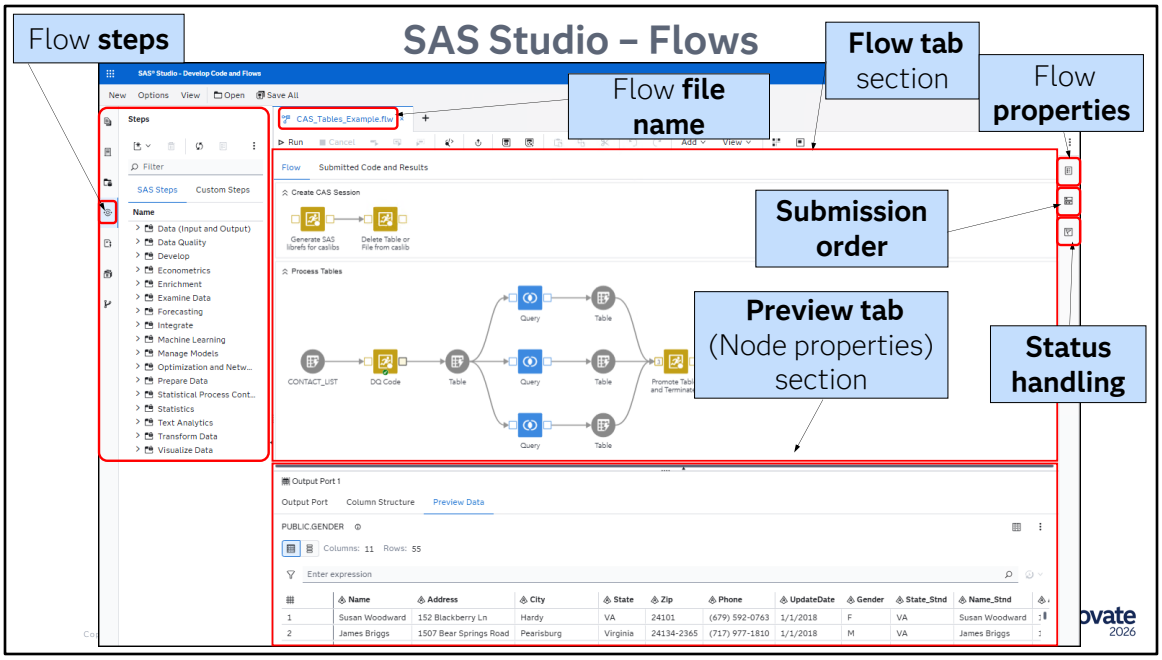
SAS Studio automatically generates SAS code for the flow. You can use flows to prepare data for reporting and analysis.

You can branch a flow as shown in this screenshot and have multiple flows in one file. Flows in separate swimlanes will run in specified execution order and flows within the same swimlane will run in parallel.

For more information, refer to the "Working with Flows" section of the SAS Studio: User's Guide.

Notes:
https://go.documentation.sas.com/doc/en/sasstudiocdc/default/webeditorcdc/webeditorflows/titlepage.htm

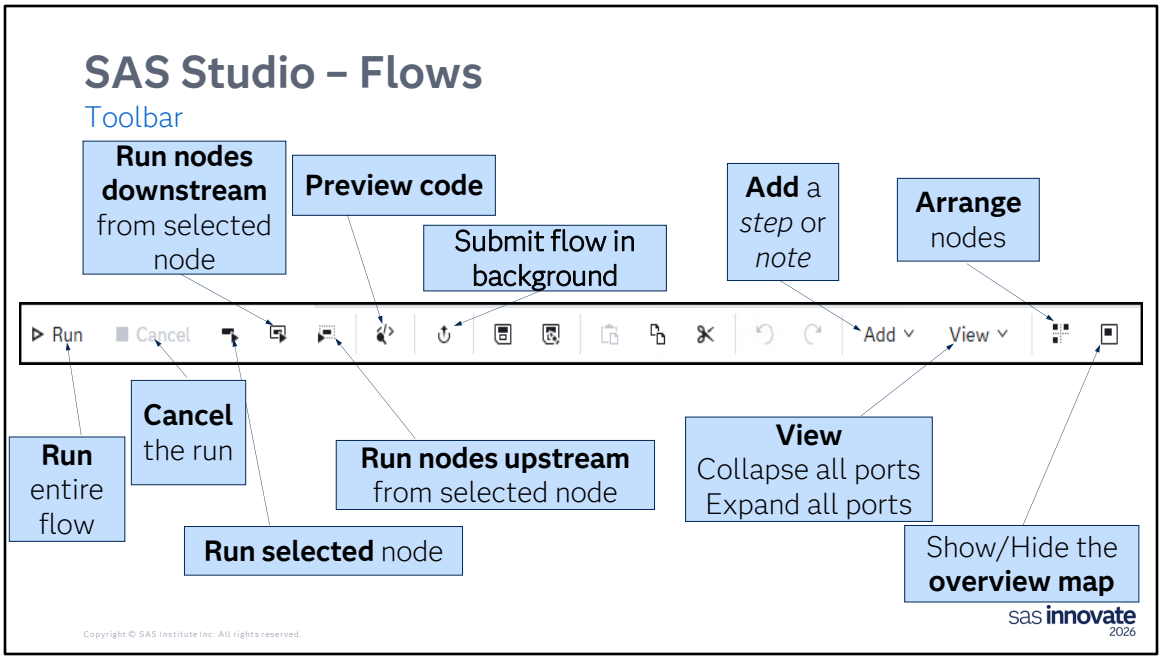
Copyright © SAS Institute Inc. All rights reserved.



Let's look at the parts of a SAS Studio Flow.

- At the top of the tab is the name of the SAS Studio Flow file.
- On the left-hand side, the Steps pane is displayed. You can add these steps to your flow canvas.
- The upper part of the main section is the Flow tab section.
- On the right-hand side of the flow tab section is the Flow properties, submission order, and status handling buttons.
- On the bottom-half of the screen is the Preview tab section also known as the node properties section.

Copyright © SAS Institute Inc. All rights reserved.



Here is the toolbar for a SAS Studio Flow. From the toolbar you can:

- Run the entire flow.
- **Cancel** the run of the flow.
- Run a single selected node in the flow.
- Run nodes upstream from selected node *does not* include the running of the selected node. Run nodes downstream from selected node *does* include the running of the selected node.
- Use this button to **preview the code** for the flow.
- Submit flow in background means that the flow can run while you continue to use SAS Studio.
- **Add** a step or note to the flow.
- Select **View** to **collapse or expand all ports** in the flow.
- You can auto-**arrange** nodes in the flow.
- Select to show or hide the **overview map** of a flow. An overview map can be useful if you have large flow with many nodes.

Copyright © SAS Institute Inc. All rights reserved.

SAS Studio – Flows

SAS Step Categories		
Data (Input and Output)	Data Quality	Develop (Code)
Econometrics	Enrichment	Examine Data
Forecasting	Integrate	Machine Learning
Manage Models	Optimization and Network Analysis	Prepare Data
Statistical Process Control	Statistics	Text Analytics
Transform Data	Visualize Data	Custom Steps

Copyright © SAS Institute Inc. All rights reserved.

sas innovate 2026

SAS Studio Flow steps are organized into categories based on the user's desired task.
Specific steps are available to:

- Work with input or output data
- Enhance data quality
- Develop code
- Work with econometrics
- Enrich data
- Examine data
- Forecasting
- Integrate data
- Manage machine learning
- Manage models
- Perform optimization and network analysis
- Prepare data
- Execute statistical process control
- Generate statistics
- Complete text analytics
- Transform data, or

Copyright © SAS Institute Inc. All rights reserved.

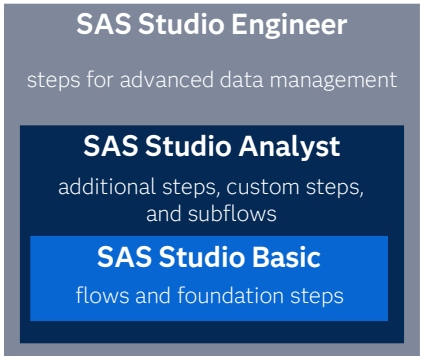
sas innovate 2026

- Visualize data.

Additionally, SAS Studio offers a separate interface for users to design completely custom steps based on their business needs.

Copyright © SAS Institute Inc. All rights reserved.

SAS Studio – Licenses



Copyright © SAS Institute Inc. All rights reserved.

SAS Studio is available in three different licenses, which vary in functionality.

SAS Studio Basic is the lowest license tier, which is designed for the general SAS user who needs foundational features only. This license includes basic steps to access, explore, transform, and export data.

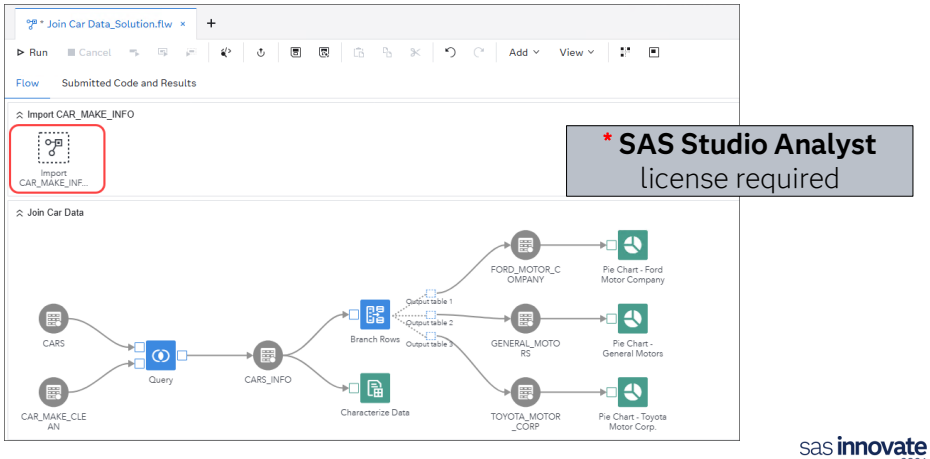
The next tier is SAS Studio Analyst, which is designed for the SAS user who needs to perform advanced analytics. This license includes intermediate steps to examine, analyze, enhance, and visualize data in addition to all features of the SAS Studio Basic license. Users can also create custom steps with SAS Studio Analyst or use existing flows as subflows.

The highest tier is SAS Studio Engineer, which is designed for the SAS user who needs to complete advanced data management tasks. This license includes advanced steps to integrate, update, clean, and parse data in addition to all features of the SAS Studio Analyst and SAS Studio Basic licenses.

Copyright © SAS Institute Inc. All rights reserved.

Subflows

To simplify preliminary or closing tasks in your flows, you can embed a subflow (saved flow) into your flow.



Copyright © SAS Institute Inc. All rights reserved.

To simplify preliminary or closing tasks in your flows, you can embed a subflow (saved flow) into your flow. A subflow is useful if you have a node or group of nodes that you want to use in multiple flows. This might include steps like importing data into SAS, starting a CAS session, terminating a CAS session, resetting your tables, and more.

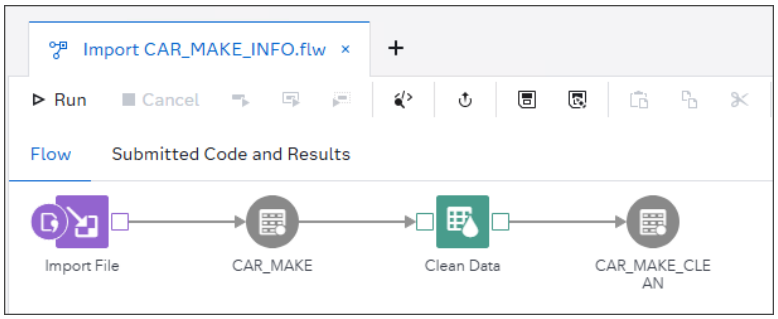
Some caveats to note about subflows are:

- You cannot run a flow that contains itself as a subflow,
- Subflows cannot be connected to other nodes in the flow, and
- The subflow feature is available only if your site licenses SAS Studio Analyst.

Copyright © SAS Institute Inc. All rights reserved.

Workshop Scenario

Objective: Make a flow that creates the **CAR_MAKE_CLEAN** table by importing a CSV file and cleaning the data. Then, use this flow as a subflow.

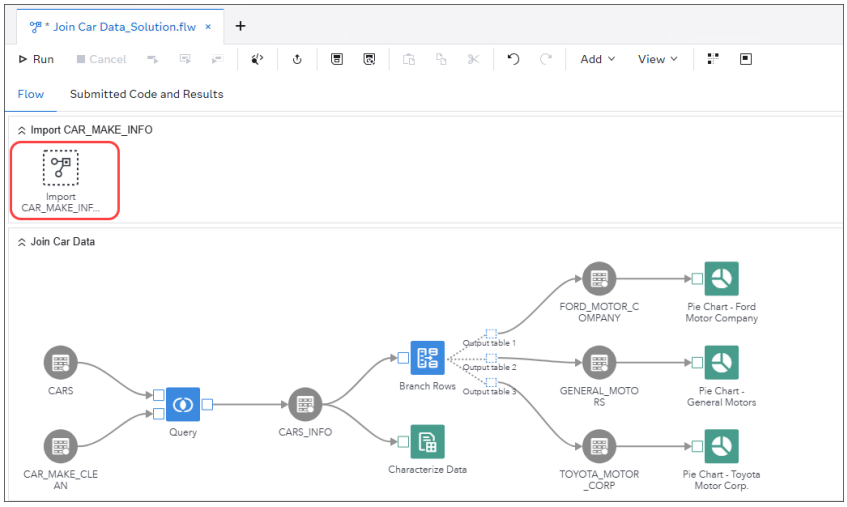


Copyright © SAS Institute Inc. All rights reserved.

The objective of this workshop is to make a flow that creates the CAR_MAKE_CLEAN table by importing a CSV file and cleaning the data. Then, use this flow as a subflow.
The initial flow will use the Import File and Clean Data steps.

Copyright © SAS Institute Inc. All rights reserved.

Workshop Scenario



Copyright © SAS Institute Inc. All rights reserved.

sas innovate 2026

The subflow will be added to an existing flow that uses the CAR_MAKE_CLEAN table. By adding this subflow, you can ensure the CAR_MAKE_CLEAN table will always be available for this flow.
This flow will use the **Query**, **Characterize Data**, **Branch Rows**, and **Pie Chart** steps.

Copyright © SAS Institute Inc. All rights reserved.

Workshop Hands-On

Subflows in SAS Viya

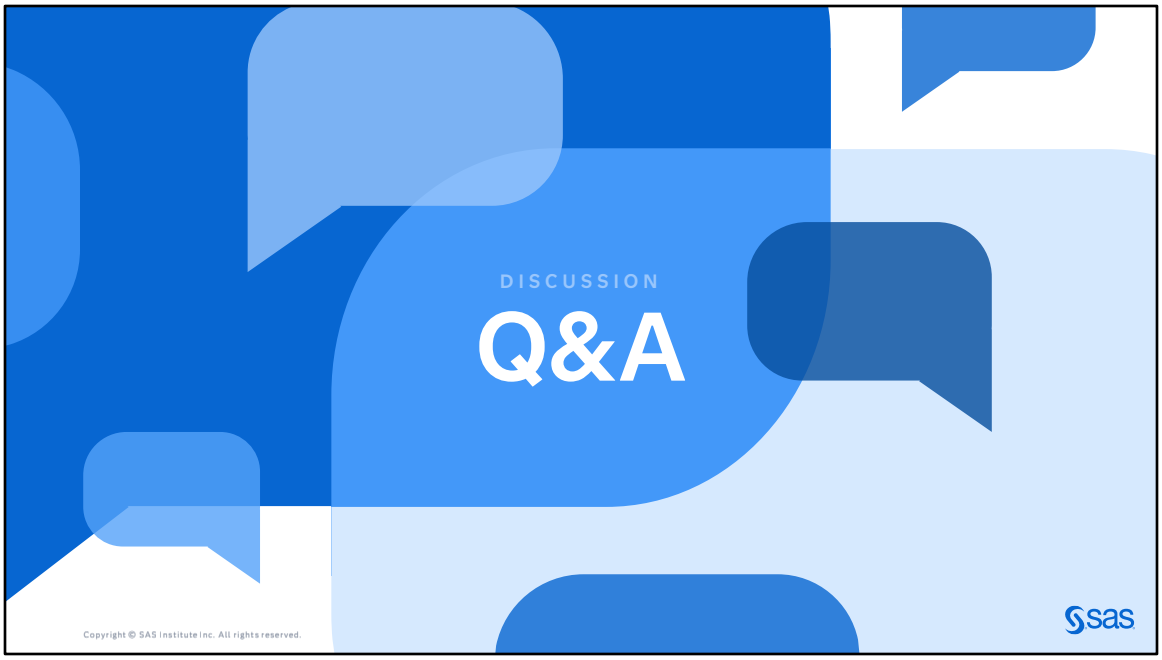
https://bit.ly/SASSubflows

Copyright © SAS Institute Inc. All rights reserved.

sas innovate 2026

Workshop instructions are available at https://github.com/SASInnovate2026/Subflows-in-SAS-Viya/blob/main/Subflows_in_SAS_Viya.md.

Copyright © SAS Institute Inc. All rights reserved.



Copyright © SAS Institute Inc. All rights reserved.

Thank you for attending!

Grace Barnhill
Grace.Barnhill@sas.com
https://linkedin.com/in/grace-barnhill

sas innovate 2026

Copyright © SAS Institute Inc. All rights reserved.

Copyright © SAS Institute Inc. All rights reserved.

sas innovate 2026